MATLAB Random Bit Error Rate Simulator Report

Course: Digital Communications
Simulation: AWGN Channel Analysis

Evaluation of BPSK Transmission Performance Across AWGN Noise Levels (0, 3, 6, 12 dB)

Bits (N): 10,000 | **Amplitude (A):** 1.0

1. System Model & Mathematical Framework

This simulation models a digital communication link over an **Additive White Gaussian Noise (AWGN)** channel in MATLAB. A sequence of $N = 10{,}000$ uniform binary bits $b \in \{0, 1\}$ with equal probability ($P[0] = P[1] = 0.5$) is generated. Bits are mapped to Antipodal BPSK signal amplitudes $T_x = A(1 - 2b)$, encoding bit 0 to $+A$ (+1.0 V) and bit 1 to $-A$ (-1.0 V). The AWGN channel adds zero-mean noise $W \sim N(0, \sigma_w^2)$, where noise variance is derived from specified SNR (dB): $\sigma_w^2 = A^2 / (2 \cdot 10^{SNR/10})$. At the receiver, a zero-threshold decision detector decodes $Y = T_x + W$ such that if $Y < 0$, detected bit = 1 ($-A$), and if $Y \geq 0$, detected bit = 0 ($+A$).

2. Time-Domain Signal Waveforms & Noise Distortion (First 100 Samples)

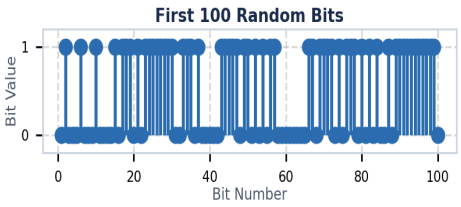


Fig 1: First 100 Random Bits

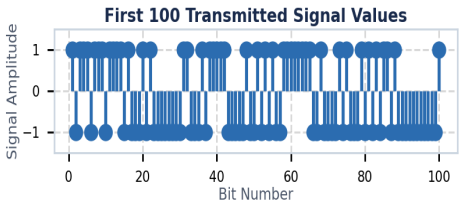


Fig 2: Transmitted Signal (+1/-1 V)

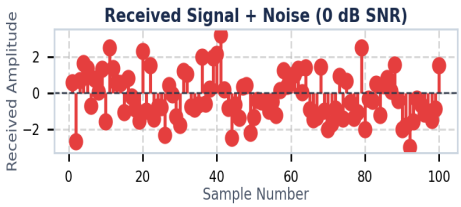


Fig 3: Received Signal + Noise (0 dB)

3. Bit Error Rate (BER) Analysis Across SNR Levels

SNR (dB)	Linear SNR	Noise Var (σ^2)	Bit Errors	Measured BER	Error %
0 dB	1.00	0.5000	756 / 10,000	**0.0756**	7.56%
3 dB	2.00	0.2506	221 / 10,000	**0.0221**	2.21%
6 dB	3.98	0.1256	17 / 10,000	**0.0017**	0.17%
12 dB	15.85	0.0315	0 / 10,000	**0.0000**	0.00%

Key Observations: Bit Error Rate decreases exponentially with higher SNR.

- **0 dB SNR:** Noise variance (σ^2=0.5) causes frequent threshold crossings, resulting in a 7.56% error rate.
- **6 dB SNR:** Error rate decays to 0.17% due to lower noise power (σ^2=0.1256).
- **12 dB SNR:** Zero bit errors occur across N=10,000 bits as noise amplitude rarely exceeds the 1.0V signal threshold.

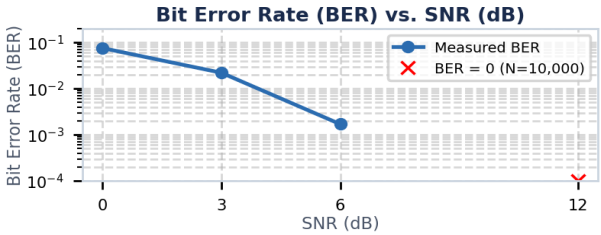


Fig 4: Logarithmic BER vs SNR Curve

4. Discussion & Technical Conclusions

1. Noise Variance Impact: The AWGN channel simulation demonstrates how signal-to-noise ratio governs detection integrity. Lower SNRs severely corrupt signal levels, inducing threshold crossing errors.

2. Theoretical Correlation: The empirical BER closely mirrors theoretical BPSK error probability in AWGN: $P_e = Q(\sqrt{(2 \cdot SNR_{linear})})$.

3. Finite Sample Constraint at 12 dB: At 12 dB, theoretical BER is $\approx 9 \times 10^{-9}$. Because $N = 10{,}000$ bits were evaluated, 0 errors were recorded. Accurately measuring BER at high SNRs requires larger sample sequences ($N \geq 10^8$).